

DATA 266 - Lab 2

Part 1: MultiModal Chatbot System (Kaggle Competition) (40 Marks)

Implement a MultiModal Chatbot system using the dataset found in the Kaggle Competition. Your chatbot must engage in multi-turn conversations about health documents, retrieve relevant information from text, figures, and tables, and generate natural, contextual responses.

Your chatbot must:

- Handle **multi-turn conversations** with context awareness
- Provide **conversational responses** (not just factual answers)
- Reference previous exchanges in the conversation
- Ask clarifying questions when needed
- Cite sources naturally (e.g., "According to Figure 5...")

1. Data Processing & Storage (6 Marks)

- Extract and process text content from PDF documents
- Extract visual elements (figures and tables) with their captions and numbers
- Generate embeddings for both text chunks and visual elements
- Store embeddings in a vector database (ChromaDB, FAISS, Weaviate, Pinecone, or similar)
- Maintain mapping between content and source figure/table numbers

2. Retrieval System with Context Awareness (8 Marks)

- Implement efficient indexing of text and image embeddings
- Build a retrieval pipeline that:
 - Accepts user query AND conversation history
 - Retrieves relevant text chunks considering context
 - Identifies top-3 relevant figures/tables
- Use ranking methods such as:
 - Contextual retrieval (query reformulation based on history)
 - Cosine similarity with conversation-aware embeddings
 - Hybrid search (combining dense and sparse retrieval)

3. Conversational Response Generation (10 Marks)

- Use a Large Language Model (LLM) to generate chatbot responses that:
 - Maintain conversation context across multiple turns
 - Reference previous messages when relevant
 - Provide natural, engaging responses (not robotic)
 - Appropriately cite figures/tables (e.g., "As shown in Figure 5...")
 - Ask clarifying questions when needed
- Implement conversation memory/context management
- Handle different types of user intents

4. Chatbot Pipeline & Integration (6 Marks)

- Create an end-to-end conversational pipeline that:
 - Tracks conversation state
 - Manages context window efficiently
 - Retrieves information for each turn
 - Generates contextually appropriate responses
 - Outputs submission file in correct format
- Include proper conversation flow handling
- Ensure reproducibility with clear documentation

Evaluation

Assess system performance using:

- **BERTScore** – Evaluates the semantic quality of generated responses
- **Token F1** – Evaluates exact token overlap of generated responses
- **Rogue-L** – Evaluates the Longest Common Subsequence (LCS) of generated responses

Metric Computation: (40%) BERTScore + (30%) Token F1 + (30%) Rogue-L

CSV Submission Format (Kaggle Leaderboard Submission)

- There are 14 input questions that your application needs to answer based on the provided document. This can be found under Files >> Labs >> Lab_2 >> Part_2 >> lab2_questions.csv
- The response should be the retrieved text and an image (set the value to 0 if there is no relevant image number for this response).
- Submit a CSV file containing the answers to all the questions in the following format:

id	conversation_id	question_id	answer	figure_references
xx	xx	xx	xxxx	xxxx

Leaderboard Rank	Points Awarded
1-5	10
6-10	8
11-15	6
15-20	5
20+	4

Part 2: Fine-Tuning Diffusion Models for Image Generation (30 Marks)

Fine-tune Diffusion models on a dataset of your choice and evaluate the generated images using Inception Score (IS) and CLIP Similarity Score.

Your goal is to:

1. Fine-tune Diffusion models (Stable Diffusion, Qwen, Flux, etc.) on a dataset of your choice.
2. Generate images based on text prompts.
3. Evaluate the quality and text-alignment of generated images using IS and CLIP scores.
4. Compare and Contrast the results of your findings and evaluate on potential reasons for those differences

Dataset Preparation & Preprocessing

- Select a custom dataset for fine-tuning (e.g., art, landscapes, anime, faces).
- If applicable, preprocess images to match Diffusion’s input format (512x512 resolution, normalized).

Fine-Tuning Stable Diffusion

- Train model (LoRA or QLoRA or full fine tuning) on your dataset.
- Use pretrained models (e.g., SD 1.5, SDXL, etc) as a base.
- Save the fine-tuned model weights for inference.

Generate Images with Text Prompts

- Run inference on at least 5 different prompts.
- Ensure high-quality and diverse outputs.

Evaluate Model Performance

Use the following **three key evaluation metrics**:

- **Inception Score (IS)** – Measures the diversity and quality of generated images.
- **CLIP Similarity Score** – Measures how well generated images align with their text prompts.
- **Human Observed Score** – Measures Quality of generated images through human observation.

Part 3: Developing an Agentic AI E-Commerce Assistant (30 Marks)

Extend an AI e-commerce assistant by implementing autonomous agents that interact with external APIs to provide a comprehensive personalized product recommendations and purchasing guidance. The agents will collaborate to retrieve information and generate a personalized shopping list based on user queries.

Implementing Agentic APIs

You will develop four agents, each responsible for fetching specific information from external APIs:

1. Product Search API Agent

- **Task:** Find products matching user preferences and requirements.
- **Requirements:** Integrate with e-commerce APIs (Amazon, eBay, etc) for product

discovery to obtain details such as product name, price, and customer reviews.

2. Price Comparison API Agent

- **Task:** Compare prices across multiple retailers/vendors and potential products
- **Requirements:** Track price history and identify best deals using price tracking APIs

3. Review Analysis Agent

- **Task:** Analyze customer reviews and extract sentiment insights
- **Requirements:** Employ an external sentiment analysis API (e.g., HuggingFace Inference API) to process review data to provide quality assessments and common feedback themes

4. Recommendation Engine Agent

- **Task:** Synthesize information from the other agents to generate a personalized Product Recommendation.
- **Requirements:** Combine search results, price data, and review insights to suggest optimal products

Integration and Execution

1. **Modular Design:** Ensure each agent functions as a modular component within the AI system.
2. **Communication Workflow:** Implement a centralized workflow to facilitate communication between agents.
3. **Testing:** Validate with various shopping scenarios (budget constraints, specific requirements, comparative shopping)